

Taipei MRT Retail Dynamics

A reproducible spatial data science comparison of Gongguan, Zhongxiao Fuxing, and Zhongshan

Group 12 | NCCU Innofest 2026 | Analytical unit: 500 m MRT station catchments



GitHub repo
code, data, maps

Research question: How do retail density, retail mix, catchment definition, and MRT passenger flow differ across three Taipei MRT station areas?

OSM POIs
2,160

Commercial and retail points inside baseline 500 m catchments.

Stations
3

Gongguan, Zhongxiao Fuxing, and Zhongshan.

MRT exits
15

Verified station-exit points used for sensitivity testing.

Passenger flow
1.36M

March 2026 entries plus exits across the three stations.

Research Frame

Core idea. MRT stations concentrate pedestrian flows and shape nearby commercial environments. This project compares three different station types using one reproducible spatial workflow.

Verified metrics. Station coordinates, 500 m buffer distance, EPSG:3826 metric projection, station exits, and March 2026 MRT passenger flow.

Proxy metrics. OpenStreetMap POI counts, category shares, circular buffers, exit-union buffers, and walk-network catchments.

Method

- Create station points from supplied coordinates.
- Project to EPSG:3826 and create 500 m buffers.
- Collect OSM POIs and classify retail categories.
- Compare circular, exit-based, and walk-network catchments.
- Add verified MRT passenger flow as a separate context metric.

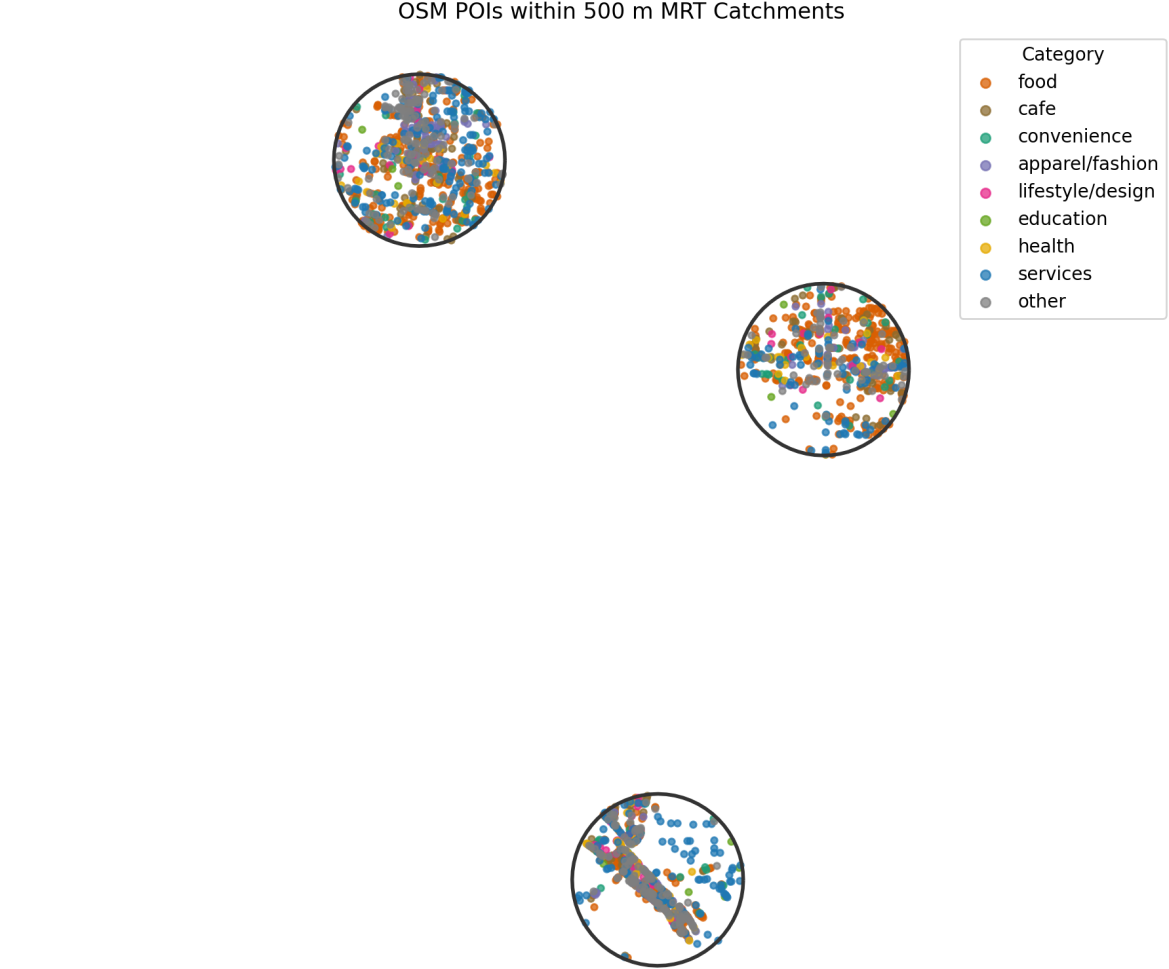
Baseline Results

Station	OSM POIs	Flow
Zhongshan	867	551,338
Zhongxiao Fuxing	561	479,242
Gongguan	732	324,978

Data Status

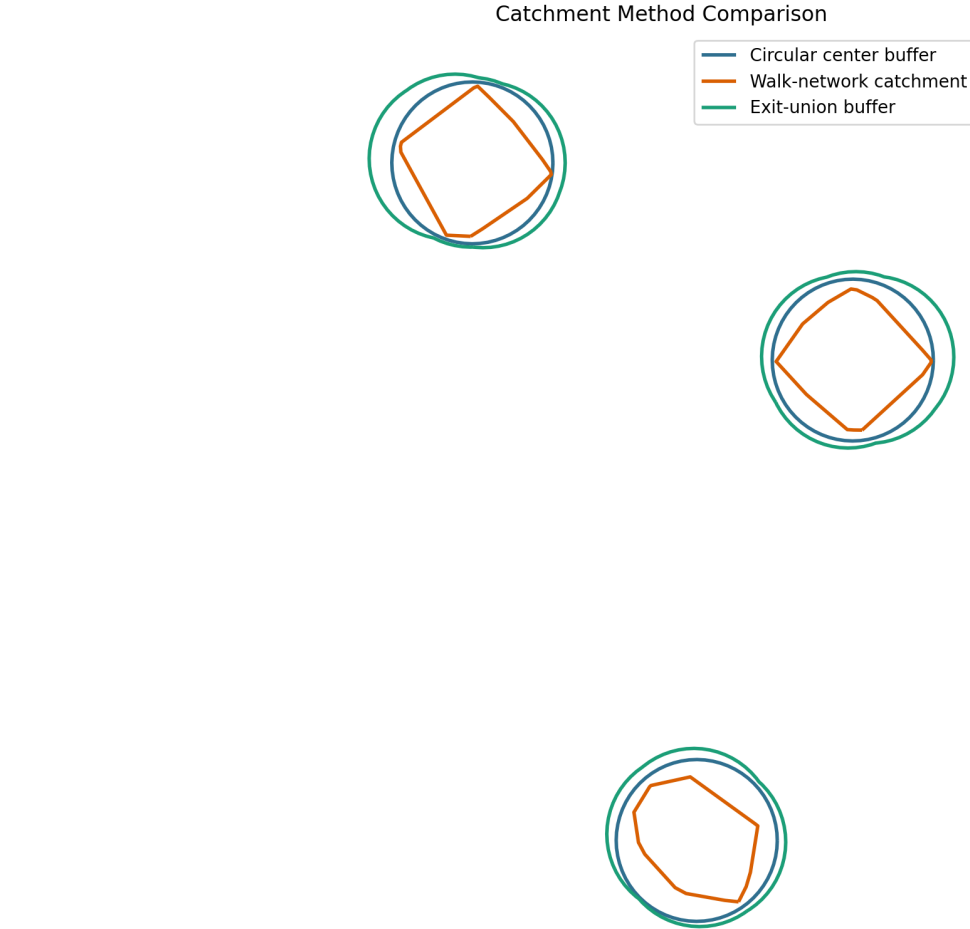
Layer	Role
Station coordinates	verified input
MRT exits	verified spatial input
MRT OD flow	verified metric
OSM POIs	proxy for retail mix
Walk network	proxy catchment

Observed Retail POIs



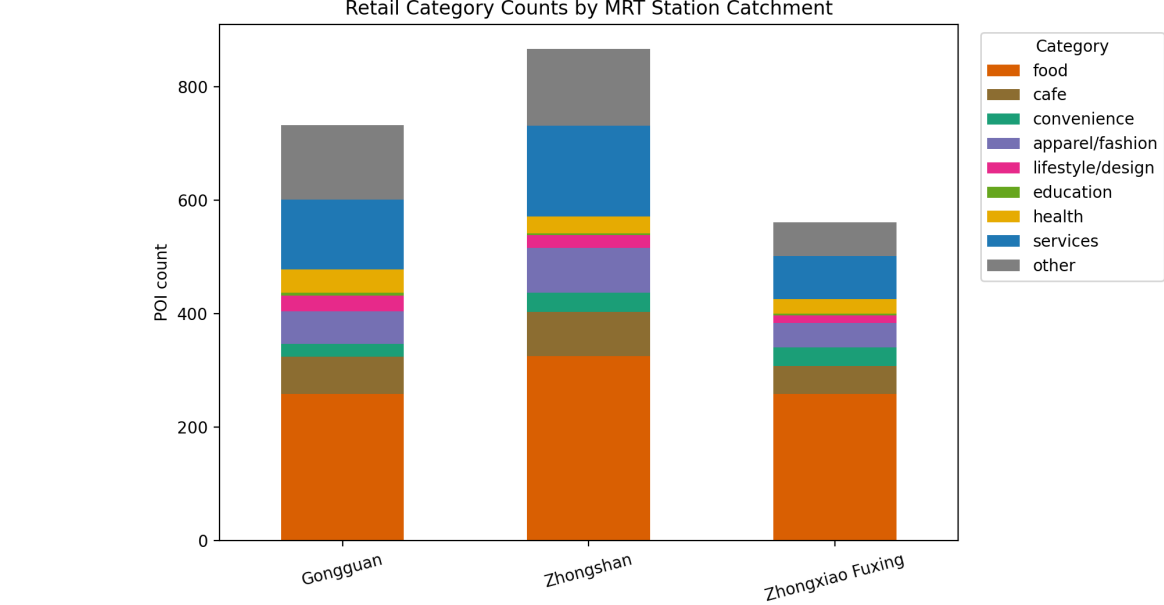
OSM-based proxy for station-area commercial density.

Catchment Sensitivity

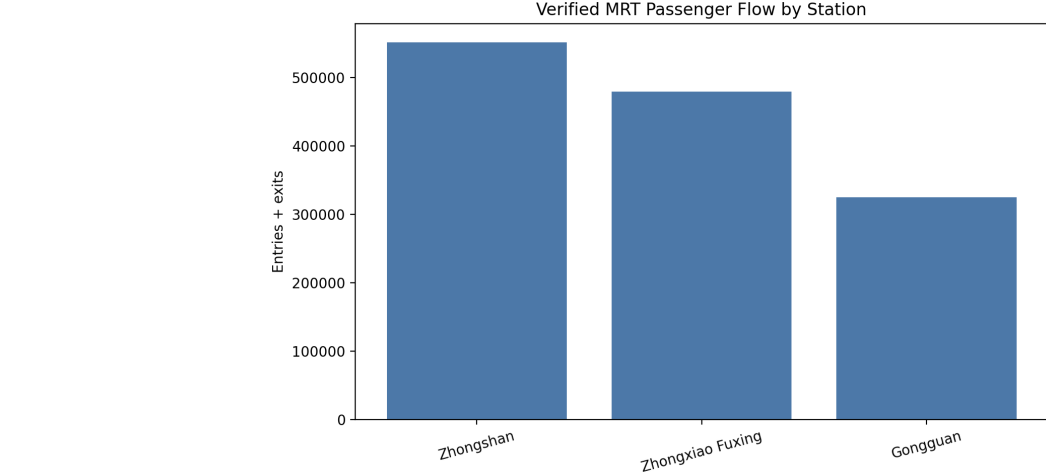


Circular, exit-union, and walk-network catchments change the observed footprint.

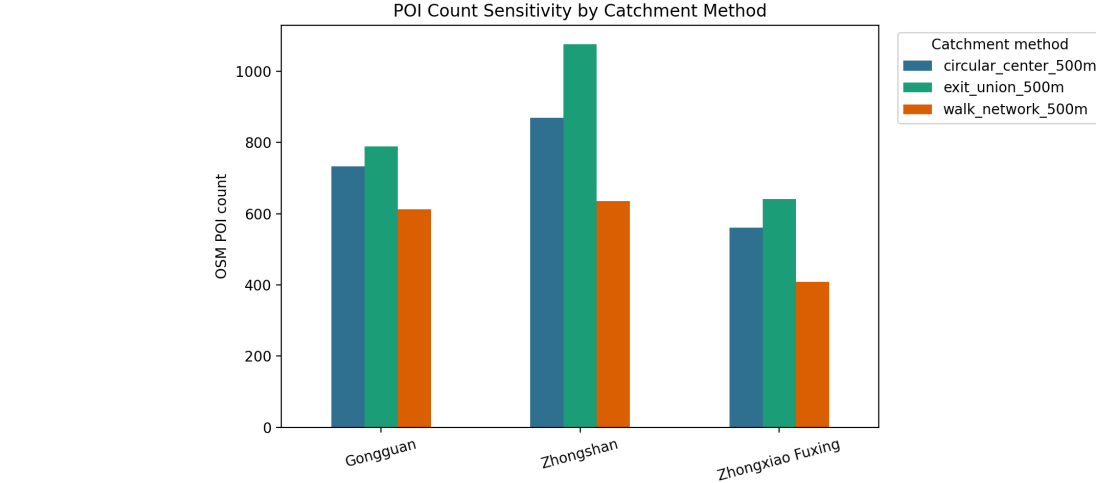
Retail Category Mix



Passenger Flow



POI Count by Catchment Method



Limitations

OSM is not an official business register. POI counts do not measure revenue, vacancy, rent, store size, or actual retail visits. MRT passenger flow measures station entries and exits, not shopping behavior. Business turnover data should only be added after transparent geocoding.

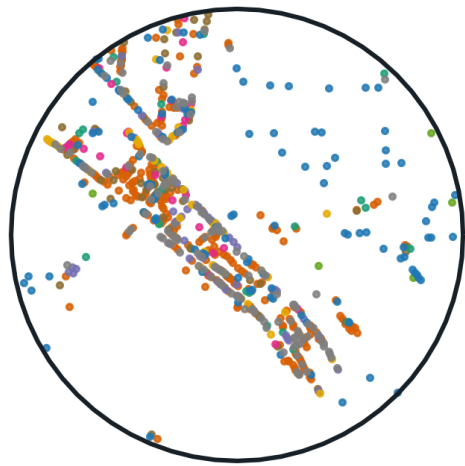
Next Step

Next: validate OSM categories manually, add land-use and building footprints, and extend the same workflow to more MRT stations.

Findings

- Zhongshan is the largest observed commercial node.** It has the highest baseline POI count and the highest verified passenger flow.
- Gongguan shows a student-oriented food and cafe profile.** High POI density remains visible despite lower station flow.
- Zhongxiao Fuxing has high flow but fewer mapped POIs.** Premium retail, malls, vertical stores, and underground activity need richer datasets than POI counts.
- Catchment choice matters.** Exit-based buffers increase counts; walk-network catchments reduce them.

Gongguan: Student Food Node



OSM POIs

732

MRT flow

324,978

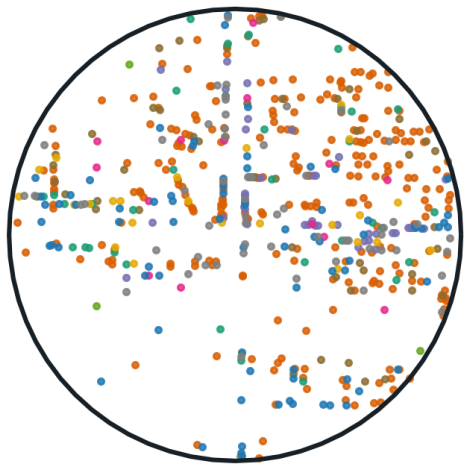
Top category

food

food, cafe, services, other

Dense student-oriented food, cafe, health, and everyday-service activity around the MRT station and university area.

Zhongxiao Fuxing: Transfer Node



OSM POIs

561

MRT flow

479,242

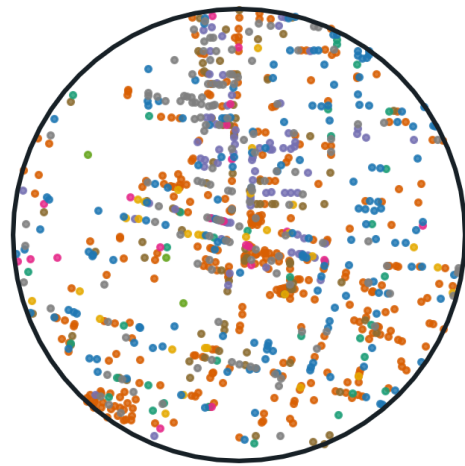
Top category

food

food, cafe, services, other

High verified passenger flow, but simple POI counts likely understate vertical malls, premium retail, and underground activity.

Zhongshan: Shopping/Culture Node



OSM POIs

867

MRT flow

551,338

Top category

food

food, cafe, services, other

Largest observed POI count and highest March 2026 MRT passenger flow among the three station catchments.

Project Deliverables

Reproducible GitHub package. 11 Python scripts, one notebook, station points, station exits, circular buffers, exit-based buffers, walk-network catchments, OSM POI GeoJSON/CSV files, comparison tables, PNG maps/charts, final Markdown report, and this A1 poster generator.

Why this matters. The project is deliberately transparent: every plotted number can be traced to a source table, and every map can be regenerated from code.